

Fig. 2: An LRC circuit with switch open

moving around two of these to create a small change, or adding new components to create a whole new version.

Options a click away. Turning the circuit on and off can be done by simply clicking on the switch. Move the mouse over the circuit, and as you pass each component, a short description of the component and its current state pop up at the lower-right corner of the screen. If you want to edit any part of the circuit, right-click on the corresponding component and you can see the edit window.

Graphs for better visibility. Coming to Scope window at the bottom of the screen, you can see instantaneous voltage, current and power as these

change with time, or you can stack these together and see a single window, and compare how the current and voltage trends vary.

As you move over the scope, the component it is graphing is highlighted in the circuit; you can change the scope data to show another component's details. You can also adjust the speed of simulation according to the speed at which you want the circuit to function—a slow movie or an action flick, it is your choice!

Beginner's luck

Let us take a look at some of the sample circuits already pre-loaded in CircuitMod. Here are a few categories

Solutions to a few errors that commonly occur

Voltage source loop with no resistance. This means one of the voltage sources in your circuit is shorted. Make sure there is some resistance across every voltage source.

Capacitor loop with no resistance. It is not allowed to have any current loops containing capacitors and no resistance. For example, capacitors connected in parallel are not allowed; you must put a resistor in series with these. Shorted capacitors are allowed.

Singular matrix. This means that your circuit is inconsistent (two different voltage sources connected to each other), or that the voltage at some point is undefined. It might also mean that some component's terminals are unconnected; for example, if you create an op-amp but have not connected anything to it yet, you will get this error.

Convergence failed. This means the simulator cannot figure out what the state of the circuit should be. Just click Reset, and hopefully that should fix it. Your circuit might be too complicated, but this happens sometimes even with examples.

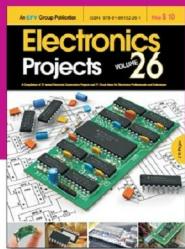
Transmission line delay too large. The transmission line delay is too large compared to the time step of the simulator, so too much memory would be required. Make the delay smaller.

Need to ground transmission line. Bottom two wires of a transmission line must always be grounded in this simulator.

(Courtesy: www.falstad.com/circuit/directions.html)

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